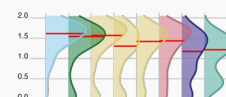
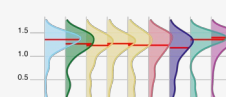
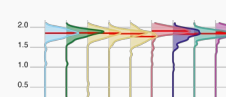
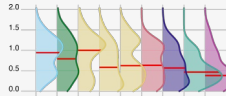
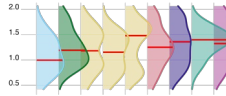
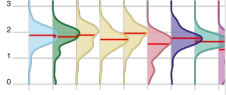
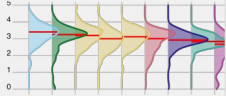
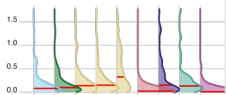
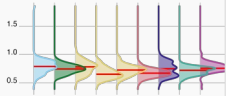
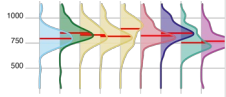
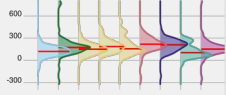
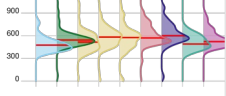
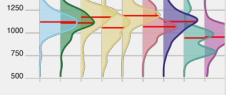
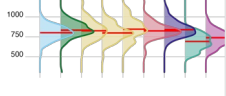


Table 3. Near-eye metrics and geographic and photoperiod context

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
	Definition/relevance	Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Duration						
Time above 1,000 lx meEDI	Bright-light exposure duration; relevant to daytime alerting and circadian entrainment.	Unit h Median 0.683 IQR 0.183 to 1.558 Participants 141 Days 816		FDR not supported; adjusted p 0.071 Part R-squared 4.7% (2.6% to 12.1%)	Estimate Ratio per 1 h: 1.206 (1.128 to 1.288) FDR supported; adjusted p <0.001 Part R-squared 9.0% (3.7% to 15.7%)	Marginal 21.5% (16.3% to 32.3%) Conditional 50.1% (40.2% to 57.6%) Participant-associated 28.6% (17.7% to 34.2%)
Time above 250 lx meEDI during wake	Waking time in recommended daytime light; relevant to alertness, entrainment, and subsequent sleep.	Unit h Median 2.417 IQR 0.850 to 4.617 Participants 141 Days 737		FDR supported; adjusted p 0.007 Part R-squared 8.6% (5.2% to 18.7%)	Estimate Ratio per 1 h: 1.136 (1.073 to 1.202) FDR supported; adjusted p <0.001 Part R-squared 6.4% (2.0% to 12.7%)	Marginal 18.0% (12.9% to 29.1%) Conditional 50.1% (41.6% to 58.0%) Participant-associated 32.1% (20.7% to 36.7%)
Time below 10 lx meEDI before sleep	Low-light time before bed; limits evening melatonin suppression and circadian delay.	Unit h Median 1.883 IQR 1.025 to 2.583 Participants 139 Days 655		FDR supported; adjusted p 0.049 Part R-squared 5.4% (3.2% to 13.4%)	Estimate Difference per 1 h: -0.134 (-0.201 to -0.066) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.2% to 10.5%)	Marginal 7.6% (4.6% to 17.2%) Conditional 38.3% (31.8% to 48.5%) Participant-associated 30.7% (21.7% to 38.1%)
Time below 1 lx meEDI during sleep	Darkness during sleep; supports nocturnal melatonin and an undisturbed sleep environment.	Unit h Median 7.133 IQR 5.967 to 8.250 Participants 141 Days 778		FDR not supported; adjusted p 0.058 Part R-squared 5.4% (3.2% to 14.2%)	Estimate Ratio per 1 h: 0.987 (0.968 to 1.006) FDR not supported; adjusted p 0.227 Part R-squared 0.6% (0.0% to 3.8%)	Marginal 8.0% (5.1% to 18.2%) Conditional 40.9% (32.8% to 48.5%) Participant-associated 32.9% (22.0% to 37.8%)

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
	Definition/relevance	Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Longest period above 250 lx meEDI	Longest sustained bright-light bout; captures continuity of daytime circadian stimulation.	Unit h Median 0.633 IQR 0.283 to 1.204 Participants 141 Days 816		FDR not supported; adjusted p 0.398 Part R-squared 2.2% (1.4% to 8.1%)	Estimate Ratio per 1 h: 1.125 (1.065 to 1.187) FDR supported; adjusted p <0.001 Part R-squared 4.9% (1.3% to 9.6%)	Marginal 10.0% (6.1% to 17.9%) Conditional 36.3% (29.9% to 45.3%) Participant-associated 26.3% (18.7% to 33.0%)
Dynamics						
Interdaily stability	Day-to-day regularity of the light–dark pattern; higher regularity supports circadian stability.	Unit dimensionless Median 0.308 IQR 0.248 to 0.380 Participants 141 Days 816		FDR not supported; adjusted p 0.228 Part R-squared 6.9% (4.1% to 20.4%)	Estimate Odds ratio per 1 h: 0.983 (0.944 to 1.024) FDR not supported; adjusted p 0.401 Part R-squared 0.4% (0.0% to 4.6%)	Marginal 15.4% (10.4% to 30.8%) Conditional 15.4% (10.4% to 30.8%) Participant-associated Not available
Intradaily variability	Within-day fragmentation of light exposure; higher values indicate less consolidated light–dark input.	Unit dimensionless Median 1.253 IQR 0.930 to 1.502 Participants 141 Days 816		FDR not supported; adjusted p 0.483 Part R-squared 5.1% (3.8% to 18.7%)	Estimate Difference per 1 h: -0.019 (-0.056 to 0.019) FDR not supported; adjusted p 0.327 Part R-squared 0.7% (0.0% to 5.7%)	Marginal 5.9% (4.6% to 21.2%) Conditional 5.9% (4.6% to 21.2%) Participant-associated Not available
Exposure history						
meEDI dose	Intensity–duration-weighted melanopic exposure; summarizes cumulative non-visual retinal light input.	Unit klx·h Median 4.960 IQR 1.936 to 12.313 Participants 141 Days 761		FDR not supported; adjusted p 0.209 Part R-squared 2.8% (1.7% to 8.6%)	Estimate Ratio per 1 h: 1.275 (1.169 to 1.391) FDR supported; adjusted p <0.001 Part R-squared 7.5% (3.1% to 12.9%)	Marginal 11.8% (7.7% to 19.7%) Conditional 33.8% (26.7% to 42.8%) Participant-associated 22.0% (14.5% to 28.7%)
Level						

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
	Definition/relevance	Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Mean melEDI	Geometric average of daily melEDI values, including zeros; summarizes overall exposure while reducing peak influence.	Unit lx Median 5.154 IQR 2.831 to 9.225 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 8.4% (4.8% to 16.6%)	Estimate Ratio per 1 h: 1.181 (1.109 to 1.257) FDR supported; adjusted p <0.001 Part R-squared 7.7% (2.9% to 13.9%)	Marginal 25.4% (18.6% to 34.4%) Conditional 57.7% (51.9% to 65.1%) Participant-associated 32.3% (24.5% to 39.4%)
Brightest 10 h geometric mean	Offset geometric mean melEDI in the brightest supported 10-hour window; describes the strength of the main sustained daily exposure.	Unit lx Median 110.566 IQR 41.513 to 243.212 Participants 141 Days 816		FDR supported; adjusted p 0.011 Part R-squared 5.7% (3.3% to 12.7%)	Estimate Ratio per 1 h: 1.243 (1.133 to 1.363) FDR supported; adjusted p <0.001 Part R-squared 5.8% (1.9% to 11.1%)	Marginal 17.6% (12.1% to 26.0%) Conditional 46.0% (39.6% to 54.3%) Participant-associated 28.4% (20.9% to 35.4%)
Darkest 10 h geometric mean	Offset geometric mean melEDI in the darkest supported 10-hour window; describes the sustained low-light period.	Unit lx Median 0.103 IQR 0.020 to 0.253 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 13.7% (9.2% to 23.9%)	Estimate Ratio per 1 h: 1.088 (1.034 to 1.144) FDR supported; adjusted p 0.002 Part R-squared 3.4% (0.5% to 8.3%)	Marginal 22.0% (15.8% to 31.8%) Conditional 61.1% (55.3% to 68.1%) Participant-associated 39.1% (30.4% to 46.2%)
Spectrum						
Melanopic daylight efficacy ratio	Mean of viable one-minute melEDI/illuminance ratios; indicates melanopic efficacy relative to visual light.	Unit dimensionless Median 0.724 IQR 0.643 to 0.795 Participants 137 Days 687		FDR supported; adjusted p <0.001 Part R-squared 9.6% (6.0% to 17.8%)	Estimate Difference per 1 h: 0.024 (0.017 to 0.031) FDR supported; adjusted p <0.001 Part R-squared 13.0% (6.7% to 20.6%)	Marginal 27.8% (21.3% to 38.1%) Conditional 60.5% (54.6% to 68.2%) Participant-associated 32.7% (24.1% to 40.0%)
Timing						
Midpoint of the brightest 10 hours	Centre time of the brightest 10 hours; indexes the main daily circadian light cue.	Unit clock time Median 13:44 IQR 12:48 to 15:00 Participants 141 Days 816		FDR supported; adjusted p <0.001 Part R-squared 6.5% (4.4% to 12.6%)	Estimate Difference per 1 h: 0.054 (-0.045 to 0.153) FDR not supported; adjusted p 0.305 Part R-squared 0.2% (0.0% to 1.8%)	Marginal 6.7% (4.6% to 13.0%) Conditional 22.9% (16.9% to 31.6%) Participant-associated 16.2% (9.6% to 22.4%)

Near-eye personal light-exposure metrics and their geographic and photoperiod context						
	Definition/relevance	Descriptive summary		Association evidence		Modelled variation
		Overall distribution	Site distribution	Overall site	Civil photoperiod	R ² summary
Midpoint of the darkest 10 hours	Centre time of the darkest 10 hours; indexes the main daily darkness cue.	Unit clock time Median 02:54 IQR 02:01 to 03:50 Participants 141 Days 816		FDR supported; adjusted p 0.020 Part R-squared 4.6% (3.0% to 10.5%)	Estimate Difference per 1 h: -0.148 (-0.245 to -0.050) FDR supported; adjusted p 0.004 Part R-squared 2.1% (0.3% to 5.6%)	Marginal 8.6% (5.9% to 15.8%) Conditional 28.2% (22.1% to 36.7%) Participant-associated 19.6% (12.6% to 25.8%)
First light timing above 250 lx meEDI	First waking bright-light exposure; morning timing can advance circadian phase and promote alertness.	Unit clock time Median 09:08 IQR 08:02 to 10:40 Participants 140 Days 727		FDR not supported; adjusted p 0.112 Part R-squared 3.8% (2.3% to 10.2%)	Estimate Difference per 1 h: -0.097 (-0.247 to 0.053) FDR not supported; adjusted p 0.227 Part R-squared 0.5% (0.0% to 2.8%)	Marginal 6.8% (4.4% to 14.4%) Conditional 31.0% (24.6% to 40.5%) Participant-associated 24.2% (16.2% to 31.5%)
Last light timing above 250 lx meEDI	Last bright-light exposure; later timing may delay circadian phase and sleep onset.	Unit clock time Median 18:08 IQR 16:27 to 19:42 Participants 141 Days 687		FDR supported; adjusted p <0.001 Part R-squared 11.6% (8.3% to 18.6%)	Estimate Difference per 1 h: 0.323 (0.186 to 0.460) FDR supported; adjusted p <0.001 Part R-squared 4.5% (1.5% to 8.5%)	Marginal 22.2% (17.0% to 30.2%) Conditional 37.9% (31.6% to 46.3%) Participant-associated 15.8% (9.7% to 22.3%)
Mean timing of exposure above 250 lx meEDI	Average bright-light timing; summarizes the phase of daily circadian stimulation.	Unit clock time Median 13:29 IQR 12:29 to 14:39 Participants 141 Days 742		FDR supported; adjusted p <0.001 Part R-squared 12.7% (9.4% to 19.2%)	Estimate Difference per 1 h: 0.121 (0.028 to 0.213) FDR supported; adjusted p 0.013 Part R-squared 1.2% (0.1% to 3.6%)	Marginal 14.7% (10.7% to 21.9%) Conditional 25.7% (19.8% to 34.2%) Participant-associated 11.0% (5.3% to 17.2%)

Overall distributions are median (interquartile range). Participants is the participant count and Days is the participant-day count. Site and photoperiod part-R² values can contain overlapping fitted information and must not be summed. Participant-associated R² is the conditional-minus-marginal contribution associated with the participant random intercept and is undefined for participant-level outcomes. The MDER uses the mean of viable minute-level ratios in both the descriptive summary and geographic-association model cells.

Table 3: Near-eye personal light-exposure metrics and their geographic and photoperiod context. Descriptive exposure values are medians with interquartile ranges (25th to 75th percentiles). Density plots show site distributions and are descriptive. Metrics are grouped into thematic streams and describe distinct features of the daily exposure pattern rather than interchangeable health-response estimates. Model cells report overall-site evidence, civil-photoperiod associations with 95% confidence intervals, FDR decisions, participant-associated model information and model-specific R² summaries where applicable. Site and photoperiod part-R² values may contain overlapping fitted information and must not be summed. Associations are observational. Absolute latitude is fixed within and confounded with study site, so separate models containing either site or latitude are reported; complete latitude and model-adequacy results are provided in the Supplementary Information.